

Class Board\_grid:

Board\_cell ;

- values like [1,100]

Board\_size():

- return m\*n  
- most appropriate  
rectangle or square.

Board\_grid():

-return list of list – grid  
format.

example:

Board\_cells = [1 to 100]

Board\_size = 10\*10

Board\_grid = [[1 to 10],[11 to  
20],...[91 to 100].

No\_of\_snakes():

- return no\_of\_snakes.

No\_of\_ladders():

-return no of ladders.

( if 100 cells  $\square$  no.of.snakes =  $100*0.1$   
and no.of.ladders =  $100*0.1$  (Must be  
int)

Class Board(Board\_grid,Snake,Ladder):

Start\_point;

- player start point.

Destination;

- player end point.

Board\_grid;

- list of list – grid format.

Snakes;

- list of list  
- list of snakes head  
and tail.

Ladders;

- list of list  
- list of ladders base and  
top.

snake\_head; - from snake class

ladders\_base; - from ladder class

position\_after\_ladder\_jump;

- From Ladder class.

position\_after\_snake\_bite;

-From snake class.

## Class Snake:

Number\_of\_snakes ;

- From Board\_grid class.

snake\_head;

- list of heads

Example : [14,21...]

snake\_head\_tail;

- list of snake head and tails. Example, [[23,2]...]

snake\_bite():

return tail of the snake

Example 23 returns 2.

## Class Ladders:

Number\_of\_ladders ;

- From Board\_grid class.

ladder\_base;

- list of base

Example : [14,21...]

ladder\_base\_top;

- list of ladders base and tops. Example, [[2,22]...]

jump\_ladder():

return top of the ladder

Example 2 returns 22.

Class Player\_elapsed\_time:

Players\_start\_time;

Players\_end\_time;

Time\_taken():

return elapsed time.

Class Player\_score(Die,Player\_elapsed\_time):

Players\_total\_die\_use; - From Die class

Players\_time\_taken; - Form Player elapsed time class.

Score():

return players score.

Example:

score = (total die use + time taken) \* 100.

Class Die:

Die range;

Die minimum value;

Die maximum value;

Total\_die\_use():

- return total no of times using die.

Class Game(Board,Player):

player\_start\_point;  
player\_destination;  
player\_die\_value;  
Board\_snakes;  
Board\_ladders;

Game():

- Timer on
- Die value accumulation
- Check current position is snake or ladder
- Calculate total number of moves.
- Score of a player
- Timer end

Return:

- Time taken
- Total no of moves
- Score

Class Player(Die, Player\_score, Player\_elapsed\_time):

Player\_name ;

- Name of the player.

player\_current\_die\_value;

- From die class.

player\_current\_score;

- From Player\_score class.

player\_current\_elapsed\_time;

-From Player\_elapsed\_time.